F1 continuation: reduced cubic-divisor membership on P-256

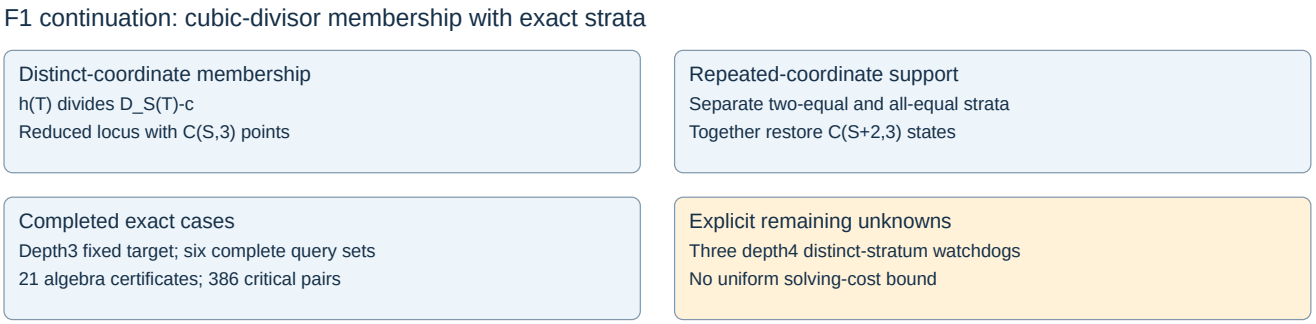

Figure 1: Editable study diagram; exact scope and evidence paths appear in the report.

2026-10-25 · TASK-20261025-13c38e · exact implementation and solver-formulation validation. This continuation completes the fixed k=3 membership+S4 target that the original forward formulation left unknown. It also proves and checks reduced membership ideals for all coordinate-multiplicity strata. Distinct-coordinate solving at k=4 remains unknown after three fresh bounded attempts. No performance trial, speed advantage, asymptotic benefit or official research transition is claimed.

Inputs, scope and representation

The original F1 package and this card were supplied as pushed commit c35b9da6a65e019eaa517fc1a5f7774428ff35db. All output is under this new tracks/f1/cubic-divisor/ directory; original evidence is unchanged. The exact published P-256 p,a,b,r and 35-term symmetric S4 are inherited from the committed baseline and original F1 source. execution.json (execution.json) binds their hashes and the source/commands used here. No Git, PR, shared-file or external service action was performed.

Use $F(t)=D_{(2^k)}(t)-c$ with $c\neq\pm2$ and the monic cubic $h=t^3-s_1t^2+s_2t-s_3$. Repeated squaring computes the three coefficients of F mod h without constructing $D_{(2^k)}$ densely. Setting them to zero means h divides F. For separable F this automatically selects three distinct coordinate roots. It avoids the earlier triple-equal terminal's scheme multiplicities, while repeated coordinates are preserved through separate systems.

The exact three-stratum construction and its proof are in derivation.md (derivation.md):

Coordinate stratum	Equations before S4	Geometric membership points
Three distinct	Three coefficients of F mod h vanish	C(S,3)
Exactly two equal (x,x,y)	F(x)=0 and $\Delta_F(x,y)=0$	S(S−1)
All equal (x,x,x)	F(x)=0	S

Here $\Delta_F=(F(x)-F(y))/(x-y)$ is a polynomial computed by a short circuit. On the diagonal it equals F'(x), so separability excludes x=y without a saturation variable. Each ideal is reduced. Joining S4 keeps a subset of a finite reduced algebra and therefore remains reduced. All tested F split completely over Fp, so all geometric points are rational. This rationality is checked rather than assumed from the field size.

Completed solver and completeness checks

The algebraic adapter expands the remainder coefficients, joins S4, constructs a grevlex Groebner basis, then performs exact lex/root extraction. Each successful basis checks original generator reductions, every critical pair, basis vanishing on every complete reference zero, and equality of finite quotient dimension with that zero count. A separate implementation of sparse modular polynomial division replays these certificates without calling SymPy. Complete geometric references are enumerated for certification; they do not feed coefficients or roots into the Groebner search. The delivered validation harness includes that exhaustive reference work and does not claim to avoid its cost.

All accepted coordinate states are factored and fully lifted with signs on P-256. The three strata's signed solutions are united and compared with a separate exhaustive signed-point triple implementation. Equal x-coordinates can lift to opposite points; this includes inverse-pair cancellations.

Fixed public query	Distinct / two-equal / all-equal S4 states	Signed point solutions	Algebraic completeness
k=2 constructed positive	1 / 0 / 0	1	Complete
k=3 original constructed positive	1 / 0 / 0	1	Complete
k=3 constructed two-equal coordinates	0 / 1 / 0	1	Complete
k=3 constructed all-equal coordinates	0 / 0 / 1	1	Complete
k=3 base-member cancellation control	0 / 7 / 1	7	Complete
k=3 published generator G	0 / 0 / 0	0	Complete empty set
k=4 original constructed positive	unknown / 0 / 0	unknown; reference has 1	Partial: distinct solver stopped

The k=3 fixed terminal and public target are the original fixture. k=2 uses its same subfibre as the earlier forward adapter. k=4 uses the committed generic k=4 fixture. These solver fixtures have actual canonical base sizes M=4,7,10, respectively; the k=2 subfibre differs from the first package's matched-M=3 coset. Exact c, targets, roots, signed bases, all accepted states and all point witness indices are retained in the result and recombination files. Constructed positives establish correctness only. The fixed G empty check is a finite-domain fact, not a uniform-target sampling result. Base cancellations are controls and are not counted as independent useful relations.

Additional exact outcomes:

- Membership-only k=3 gives a reduced quotient of dimension 56=C(8,3), 28 basis

polynomials and 378 checked critical pairs. These are exact algebraic counts.

- Across 21 successful algebraic attempts, the separate sparse verifier checks

386 critical pairs. All 21 certificates pass.

- Complete linear factorization, divisor identities, nonzero Jacobians and the

equality det Jacobian=Res(h,F/h) are checked at all 4+56+560 distinct divisors for S=4,8,16 over the actual field.

- Forty deterministic actual-field coefficient cases compare symbolic remainder

circuits with independent polynomial long division and automatic derivatives.

- Seven three-distinct divisor/remainder/nonzero-Jacobian witnesses reach depths

k=6,7,8,16,32,64,96. They certify representations, not complete high-depth solves.

- The out-of-scope actual-field control F=D8−2, $h=t^3-4t$ divides F but has zero

remainder Jacobian determinant. This checks that separability is a substantive hypothesis for the proof; it does not classify every nonseparable fibre.

Preserved unknown attempts and costs

attempts-01/receipt.json (attempts-01/receipt.json) retains 22 sequential attempts: 21 complete, one stopped. The default SymPy Buchberger k=4 distinct-positive attempt stops during basis construction. attempts-02/receipt.json (attempts-02/receipt.json) retains two fresh F5B attempts, for that k=4 positive and the fixed generator; both stop during basis construction. Each has a declared 30-second watchdog and 2 GiB process address-space cap. Native stdout/stderr remain per attempt. No stopped case has an empty-set or completeness label. The original F1 watchdog is left unchanged in its original directory.

No timing or peak-memory benchmark was collected, and no adapter comparison is inferred. Environment: the pinned /workspace/dickson-solver/.venv/bin/python, CPython 3.12.14, SymPy 1.14.0, python-flint 0.8.0. The inherited native Codex runtime/model binding remains unprobed; same-session replay is not independent scientific review.

With fixed h-coefficient products shared, a uniform remainder step has 3 field squarings, 9 general multiplications, 3 small-constant scalings and 8 additions. This is an arithmetic-circuit count. Polynomial expansion still grows:

k	S	Expanded coefficient degree	Nonzero terms (R0,R1,R2)
2	4	2	2,2,3
3	8	6	10,13,14
4	16	14	67,85,81
5	32	30	496,575,546

The generic cubic-divisor quotient has length C(S,3), versus S^3 for the earlier symmetric-terminal representation; the full reduced union has C(S+2,3) states. These representation facts do not bound solver runtime, certificate-generation cost, target success, or total ECDLP complexity. The current algebraic adapter expands the input; the circuit code and deep witnesses keep the short formulation available for further work.

Artifacts and reproduction

- checks-01/receipt.json (checks-01/receipt.json): six complete query unions and

one partial unknown; individual files contain full witnesses.

- checks-01/identities.json (checks-01/identities.json): exact factor, Jacobian,

circuit, gradient, hypothesis-control and deep-witness results.

- basis-replay-01.json (basis-replay-01.json): independent-code replay of 21 bases.

- execution.json (execution.json): source/input identities, exact commands and

environment; native logs remain alongside their receipts.

- packed-artifacts.json (packed-artifacts.json): the 71,029-byte native membership

certificate is retained losslessly as gzip with its original byte hash. Its receipt still binds the decompressed bytes. No data was omitted.

- diagram.svg (diagram.svg): editable vector explanation; aggregate PDF rendering

belongs to the Coordinator.

For a fresh single-case replay from the repository root, choose a previously unused output path:

```
TRACK=research/p256_factor_base_parallel_20261025/tracks/f1/cubic-divisor
PYTHONDONTWRITEBYTECODE=1 /workspace/dickson-solver/.venv/bin/python "$TRACK/solve.py" \
  --case "$TRACK/../checks-02/generic-k3.json" --depth 3 \
  --stratum distinct --target positive --out "$TRACK/replay-distinct-k3-new.json"
```

The two supervisor scripts and check script deliberately refuse existing result directories; preserve the delivered records. verify_basis.py reads either native JSON or its exact gzip representation. Source inputs and frozen receipt hashes are retained, including the stopped attempts' reconstructible equations.

Remaining obligation

The nonreduced generic membership obstacle now has an exact replacement on all three strata. The separate obstacle is a certified global elimination method and cost bound for the unexpanded circuit. problem.md (problem.md) records that change and the next discriminating step: implement incremental elimination on the multiplication-gate circuit and replay unchanged k=3 and k=4 positive/empty controls, retaining complete or unknown outcomes. Relation rank, labels, setup, linear algebra and final target attachment remain separate ECDLP obligations.

No shared graph or problem index was edited. The Coordinator should link this progress into the original F1 problem and include the new report/diagram in the aggregate PDF. Neither filing nor archiving this package changes a hypothesis.